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Diffusion NMR of Poly(acrylic acid) Solutions: Molar Mass Scaling and pH-Induced Conformational Variation

*Arthur Lenoch, Marcel Schumacher, André H. Gröschel, Cornelia Cramer, and Monika Schönhoff**

Self-diffusion of poly(acrylic acid) (PAA) in dilute solution is studied by pulsed field gradient (PFG)-NMR, also known as DOSY, and correlated to molecular weight with scaling laws at varying pH. Fitting a diffusion coefficient distribution model to PFG-NMR data, the molecular weight polydispersity of commercial PAA samples is quantified. Due to good agreement with GPC data, diffusion measurements offer a viable alternative to estimate chain length properties with minimal sample preparation. Similar results are achieved by dynamic light scattering (DLS), however, with lower precision. The evolution of the scaling exponent with increasing pH indicates a transition from a globular state in highly acidic conditions ($\nu = 0.46$) to a partially stretched conformation with a maximum at pH 5 ($\nu = 0.67$). At higher pH, counterion screening counteracts chain expansion, while an unexpected increase in the width of the diffusion coefficient distribution occurs. A universal scaling law of the diffusivity applies for a chain length range of $25 \leq N \leq 1585$, albeit with distinctly different scaling exponents. While opposing arguments are brought forward in recent literature concerning the pH-dependence of short-chain conformations, analysis supports the view that chain conformation is significantly affected by pH, even for short chains.

1. Introduction

Identifying universal relationships in the interplay between dynamic properties and molecular weight plays a key role in our understanding of polymer solutions. An inherent property which must always be taken into account when investigating physical properties of macromolecules is their polydispersity.[1] In fact, a variety of experimental methods have been developed to not only determine the average molecular weight M_w itself, but also to specifically quantify the polydispersity of synthetic polymers. Among these, gel permeation chromatography (GPC),[2] also known as size exclusion chromatography (SEC), and dynamic light scattering (DLS)[3] have become the most prominent and widely available tools. Molecular weight measurements of poly(acrylic acid) (PAA), a weak, hydrophilic polyelectrolyte are either conducted directly with an aqueous mobile phase GPC[4–6] or with an organic solvent-based GPC after a prior esterification step to increase solubility,[7–9] as comprehensively discussed by Lakíc et al.[10] This preceding functionalization commonly consists of a methylation using hazardous reagents such as diazomethane or trimethylsilyldiazomethane.[10] This issue was addressed in a recent work by Li et al. who demonstrated that 1,1,3,3-tetramethylguanidine can effectively promote the esterification of polyacrylates and polymethacrylates with a variety of halogenated compounds.[11]

Diffusion NMR techniques have been suggested as an alternative method to study the molecular weight and chain conformation of polymers, as a multitude of characteristics like chain conformation, polymer-solvent interaction or chain length can be inferred from the self-diffusivity of macromolecules.[12]

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Compared to GPC, pulsed field gradient NMR (which is also referred to as DOSY)[13] can spectrally resolve complex mixtures and is thus relatively insensitive towards dust or impurities.[14] Additionally, NMR requires far less solvent and experimental time and puts no restriction on the polarity of the solvent or polymer concentration, as long as sufficient signal intensity is detected.[17] PFG-NMR/DOSY is nowadays a well-established experimental method to study the diffusion of small molecular species in complex mixtures,[18,19] where recent advances in the development of pulse sequences have made it possible to greatly improve the sensitivity of DOSY experiments even in case of low concentrations.[15,16] The method is, however, met with some unique challenges when polydisperse polymers come into play.[20] The question of how the presence of polydispersity affects self-diffusion measurements has been therefore addressed quite frequently in the literature, resulting in dedicated reviews[21,22] and several case studies on a variety of both synthetic[14,23–30] and biological[31–34] macromolecules.

As a particularly interesting case, polyelectrolyte diffusion and conformation have specifically received attention in both experimental[35–38] and simulation-oriented[39–42] studies, resulting in a variety of works studying PAA as a model system. Compared to strong polyelectrolytes, such as the frequently studied NaPSS,[43–47] the presence of ionizable groups across the chain lets weak polyelectrolytes like PAA exhibit complex dissociation equilibria, for which simulation-specific aspects have recently been reviewed.[42] This allows to control the conformational properties of weak polyacids by changing extrinsic conditions such as pH or ionic strength.[48] A peculiar result has been reported by Swift et al.:[35] based on fluorescence lifetime spectroscopy and diffusion NMR results, it was postulated that a molar mass limit $M_n < 16.5$ kDa exists, below which shorter PAA chains do not exhibit conformational changes upon changes in pH. In spite of an older computational study by Laguecir indeed indicating a more pronounced dependence of diffusivity for longer chains,[41] newer results by Dolce et al. refute the point made by Swift et al. by clearly demonstrating short-chain diffusion to be affected by pH, albeit to a lower degree.[36]

Here, we first explore the application of PFG-NMR experiments to acquire molecular weight distributions of protonated PAA. This procedure is based on the identification of an underlying diffusion coefficient distribution, which can be extracted from the multiexponential intensity decay. Based on the observed scaling law between M_w and the diffusion coefficient, we establish a calibration procedure for determining M_w as well as polydispersity indices of commercial PAA samples and compare them to results obtained from conventional GPC. Additionally, we investigate polyelectrolyte-specific effects of PAA chain conformations by analyzing the evolution of scaling exponents and diffusional polydispersity for increasing pH. By gaining a comprehensive picture of scaling behavior for a wide range of chain lengths at various degrees of charge, we examine whether the hypothesis of short PAA chains lacking any conformational response to pH changes can be substantiated.

2. Theoretical Background

In a typical PFG-NMR experiment, application of magnetic gradient pulses causes a spin ensemble to gain a position-dependent phase shift, which then gradually loses its coherence through diffusional displacement. This is reflected in an attenuation of the signal intensity in a spin echo experiment (**Figure 1**a), as explained in detail in several overviews on the topic.[13,18,19] More specifically, use of the pulsed field gradient stimulated echo sequence (PGSTE) has become commonplace in order to account for fast T_2 relaxation (Figure 1b). For systems with a single self-diffusion coefficient D, normalized signal decay curves obtained in a PFG-NMR experiment are well described by the Stejskal-Tanner equation:[18]

$$I/I_0 = \exp\left(-\gamma^2 G^2 \delta^2 \left(\Delta - \frac{\delta}{3}\right) \cdot D\right) \tag{1}$$

in which γ represents the nuclear gyromagnetic ratio, G is the gradient strength, δ is the gradient pulse duration, and Δ is the diffusion time. While the assumption of a single diffusion coefficient is valid in systems of small molecules in solution, it quickly breaks down if the diffusing species exhibits a significant degree of polydispersity.[13] A "bent" decay curve in an $\ln(I/I_0)$ versus k plot, as often observed for the diffusion of polymers,[20] points to an underlying distribution of chain lengths. Inferring the shape of this distribution from the original multiexponential decay data is an ill-posed inversion problem that arises frequently in a variety of fields dealing with complex exponential relaxation phenomena,[49] including dynamic light scattering,[50] rheology[51,52] or NMR relaxometry[53,54] among many others.

In order to generalize the Stejskal-Tanner Equation (1), a distribution of diffusion coefficients $P(D)$ has to be considered in the form of an integral equation[13]:

$$\frac{I}{I_0} = \int_0^\infty P(D) \exp(-kD)\ \mathrm{d}D \tag{2}$$

which formally corresponds to a Laplace transform of $P(D)$. While the choice of any particular $P(D)$ is in principle arbitrary, a reasonable function in the context of polymer diffusion is the log-normal distribution[18,20]:

$$P(D) = \frac{1}{\sqrt{2\pi}\sigma_D D} \exp\left(-\frac{[\ln D - \ln\langle D\rangle]^2}{2\sigma_D^2}\right) \tag{3}$$

which is parameterized in terms of a median diffusion coefficient $\langle D\rangle$ and a standard deviation σ_D. For computational simplification, a change of variables in the integral from Equation (3) proves to be useful:

$$\frac{I}{I_0} = \int_{-\infty}^{\infty} P^*(D) \exp(-kD)\ \mathrm{d}\ln(D)\,. \tag{4}$$

Here, a modified distribution $P*(D) = D \cdot P(D)$ is used instead, in which ⟨D⟩, in contrast to Equation (3), now now corresponds to the maximum of the distribution. Several other approaches to model polydisperse exponential decay have been discussed in the literature. For example, the classic Kohlrausch–William–Watts type stretched exponential function $I/I_0 = \exp(-[k\langle D\rangle]^\beta)$ is regularly used for describing experimental data.[13] Even though the stretched exponential does not reflect a true distribution of D in a mathematical sense, attempts to find empirical correlations between the stretch parameter β and σ_D

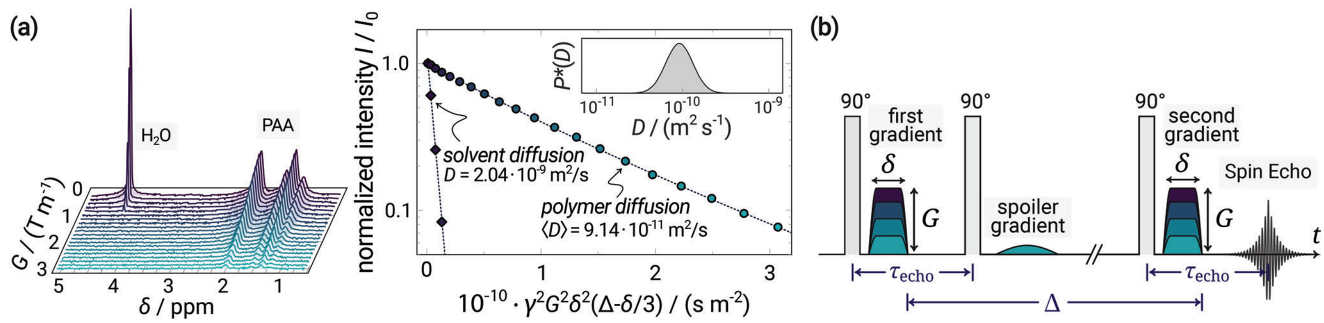

Figure 1. a) ^{1}H-NMR spectra of PAA in aqueous solution for increasing gradient strength G. An appropriate model is fitted to the PAA signal decay data, shown here in a representative Stejskal-Tanner plot, by assuming a distribution of diffusion coefficients, as depicted in the inset. b) Scheme of the PGSTE sequence with relevant experiment times.

have been made.[28] Röding et al. have suggested in a series of publications to use a gamma distribution to model $P(D)$, as the integral in Equation (3) can be analytically solved and does not need to be evaluated numerically for the fitting procedure.[12,55,56] Conveniently, the resemblance between the log-normal distribution and the gamma distribution is still sufficiently high, which makes this approach especially useful for modeling computationally challenging multimodal distributions.[12] Whereas the previously discussed parameterized fitting methods rely on some underlying specific distribution function, which is determined by nonlinear optimization algorithms, inverse Laplace transform (ILT)-based numerical methods are based on a nonparametric approach to find a smooth $P(D)$ curve of arbitrary shape. Owing to the popularization of the CONTIN algorithm,[57] ILT has become a well-established tool in diffusion NMR[24,25,58] and is a standard procedure in the processing of DOSY experiments.[59] Although ILT results may appear smooth due to being stabilized by regularization procedures,[58] several artifacts such as spurious or artificially broadened/split peaks may arise due to the inherent ill-posed nature of the problem.[60] These shortcomings must therefore be kept in mind when carelessly using ILT as a "black box" tool for analyzing data with low signal-to-noise ratios, as encountered for highly dilute polymer solutions. ILT is nevertheless particularly useful for systems displaying an underlying multimodal parameter distribution over a range of several orders of magnitude.[13]

3. Results and Discussion

3.1. Molecular Weight Determination by GPC

In order to accurately evaluate the scaling behavior of the diffusion coefficient, molecular weights of different commercial PAA samples need to be determined precisely. For this purpose, PAA was first modified with hydrophobic benzyl groups to make the polymer soluble in dimethylacetamide (DMAc) for further analysis by gel permeation chromatography. It was demonstrated by Li et al., that the esterification of PAA with a wide range of halogenated compounds can be efficiently promoted by using 1,1,3,3-tetramethylguanidine as a strong organic base, thus bypassing the need for hazardous methylation reagents.[11] We adopted a slightly modified synthetic procedure to obtain poly(benzyl acrylate) (PBzA) by reaction of different PAA samples with benzyl

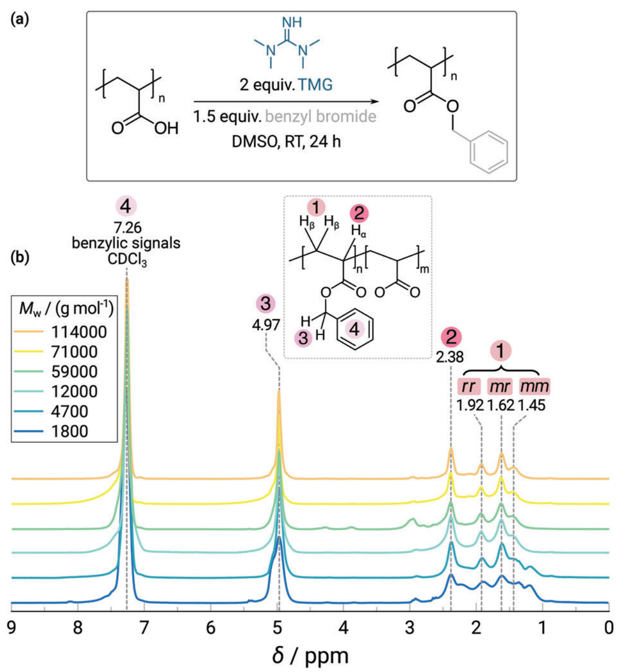

Figure 2. a) Reaction scheme of the TMG-mediated esterification of poly(acrylic acid) with benzyl bromide to poly(benzyl acrylate). b) ^{1}H-NMR spectra of poly(benzyl acrylate) samples in $CDCl_3$. The assignments *rr*, *mr*, and *mm* correspond to the respective iso, hetero-, and syndiotactic triads of backbone protons.

bromide (**Figure 2**a). ^{1}H-NMR spectra of PBzA show the appearance of additional resonances no. (3) and (4) (Figure 2b) and thus demonstrate successful benzylation of the polymer.

The superimposed spectra were deconvoluted with a set of Lorentzians to accurately determine the degree of benzylation f_{Benz}. This quantity is obtained by relating the integrated signal of the benzylic methylene group at 4.97 ppm to the signal of backbone α-protons at 2.38 ppm[11]:

$$f_{\mathrm{Benz}} = \frac{I_{\mathrm{Ph-CH_2}}/2}{I_{\mathrm{H}_\alpha}} \tag{5}$$

Degrees of benzylation as determined from ^{1}H-NMR spectra are in the range between 85 % and 94% (summarized in the Sup-

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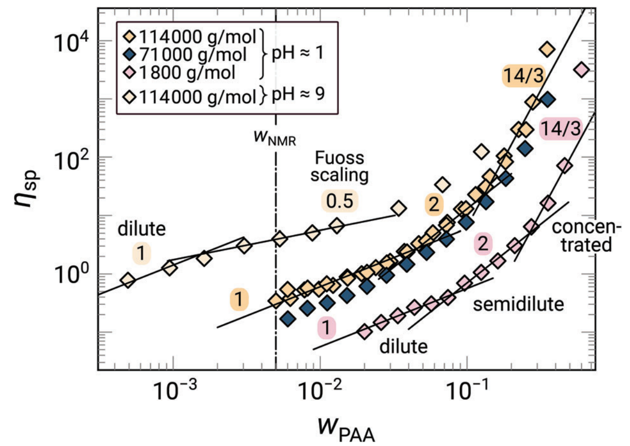

Figure 3. Polymer mass fraction dependence of the specific viscosity for three PAA samples with different M_w in 0.1 M HCl (pH ≈ 1). Expected scaling regimes and their exponents for a θ-solvent according to Colby[61] are indicated. Additionally, η_{sp} is shown for the fully neutralized PAA with the longest chain length (pH ≈ 9), where scaling regimes for a polyelectrolyte without salt are indicated. The line at w_{NMR} refers to the weight fraction of 0.5 wt.% used in NMR experiments at acidic pH.

porting Information, Table S1). Using f_{Benz}, an averaged molecular weight $M_{copoly} = f_{Benz} M_{PBzA} + (1 - f_{Benz}) M_{PAA}$ of the PBzA-*co*-PAA samples can be calculated, from which GPC curves of PBzA were rescaled by M_{PAA}/M_{copoly} to recover the molecular weight distribution of the original PAA. Parameters of the distribution (M_w and $Đ$) are given in Table S1, Supporting Information. The rescaled distribution serves as a reference distribution for the calibration of the subsequent diffusion experiments, which were all performed on the original PAA in deuterated aqueous solutions.

3.2. Diffusion NMR

3.2.1. Diffusion Coefficient Distributions and Chain Conformation in Acidic Conditions

With exact knowledge of M_w, the scaling relation between diffusion of the undissociated polyelectrolyte and its molecular weight can now be investigated. For the NMR measurements, PAA was dissolved in 0.1 M DCl in D_2O, which resulted in dilute solutions with an apparent pH* ≈ 1. At such acidic conditions, all acid sidegroups are fully protonated and the polymer should exhibit the statistics of uncharged chains. To ensure free self-diffusion of the chains without any possible overlap, the polymer weight fraction w_{PAA} for NMR experiments at acidic conditions was set to $w_{PAA} = 0.5$ wt.% We verified that this concentration lies in the dilute regime by viscometric measurements of three representative PAA samples ($M_w \approx 1\,800$, 71 000, 118 000, all in g mol^{-1}; see **Figure 3**). The specific viscosity $\eta_{sp} = (\eta_0 - \eta_s)/\eta_s$, with η_s being the solvent viscosity and η_0 the zero shear viscosity, lies well within the dilute region at w_{PAA}, as confirmed by the characteristic $\eta_{sp} \sim w_{PAA}^1$ scaling.[61]

We measured the NMR signal decays for the investigated aqueous PAA samples using pulsed-field gradient-stimulated echo experiments for at least 3 independently prepared solutions with different maximum gradient strengths, respectively. Parameters of a log-normal distribution (Equation (3)) were optimized to obtain a decay function (Equation (4)) which could describe the combined data from each measurement adequately (see **Figure 4**a). The distinctly nonlinear intensity decay indicates a significant degree of polydispersity. Fitting with the log-normal distributed exponential decay model thus results in excellent agreement, which we further highlight in the supporting information, Figure S1: neither a single exponential nor a stretched exponential model describes the data nearly as good.

In order to relate the observed diffusion coefficients (see Table S2 for fit parameters) to the molecular weight, the validity of the universal scaling law:

$$D = K M_w^{-\nu} \tag{6}$$

can be tested for the investigated system. Here, K is a system-specific constant and ν is a scaling exponent. Knowing ν provides valuable information about the chain conformation and thus about the solvent quality. Taking θ-solvents for example, a value of $\nu = 0.5$ is expected for Gaussian chains, while an improvement of solvent quality will lead to higher values of ν. In the good solvent limit, excluded volume effects dominate and $\nu = 0.588$,[61] whereas ν would approach 1 for a rigid rod chain.[62] We note that Equation (6) theoretically holds for the inverse radius of gyration R_g^{-1} rather than the diffusion coefficient, and the relation between D and R_g might involve complex hydrodynamic processes; nevertheless diffusion measurements of the hydrodynamic radius $R_H \sim D^{-1}$ have been successfully employed to yield scaling exponents, for example $\nu = 0.57$ for poly(ethylene oxide) in water.[63]

As shown in Figure 4c, a scaling exponent of $\nu = 0.46$ is found to describe the relationship between M_w and $\langle D \rangle$ very well. Notably, the statistical properties of the chain remain universal throughout a wide range of chain lengths ($25 \leq N \leq 1585$). Since $\nu < 0.5$, we can classify 0.1 M HCl as an apparently poor solvent for PAA, in which the chains thus adopt a slightly collapsed conformation. This result is surprising at first since PAA side groups are expected to retain good hydrogen bonding capabilities to surrounding water molecules. It is well-known, however, that PAA monomers interact with each other by forming intrachain hydrogen bonds.[64,65,48] Accordingly, PAA takes on a slightly globular conformation in acidic solution since intrachain hydrogen bonding is expected to increase the strength of monomer-monomer interactions. While MD simulations of short PAA chains ($20 < N < 110$) by Mintis et al. indeed predict a collapsed conformation for completely undissociated PAA ($\alpha = 0$), a remarkably low scaling exponent of $\nu = 0.25$ was found.[39] This theoretical scaling relation is however at odds with both our and another published experimental work,[36] which reported $\nu = 0.49$ for pH = 2.

Using fit functions based on the log-normal distribution further allows us to quantify the diffusional polydispersity of the system in terms of the standard deviation σ_D of $P(D)$. A direct comparison between molecular weight distributions obtained from GPC and NMR can be made if the log(D) axis is rescaled according to Equation (6). Excellent agreement between the two independently measured distributions is found (Figure 4d). By fitting a log-normal distribution to the GPC traces, the result-

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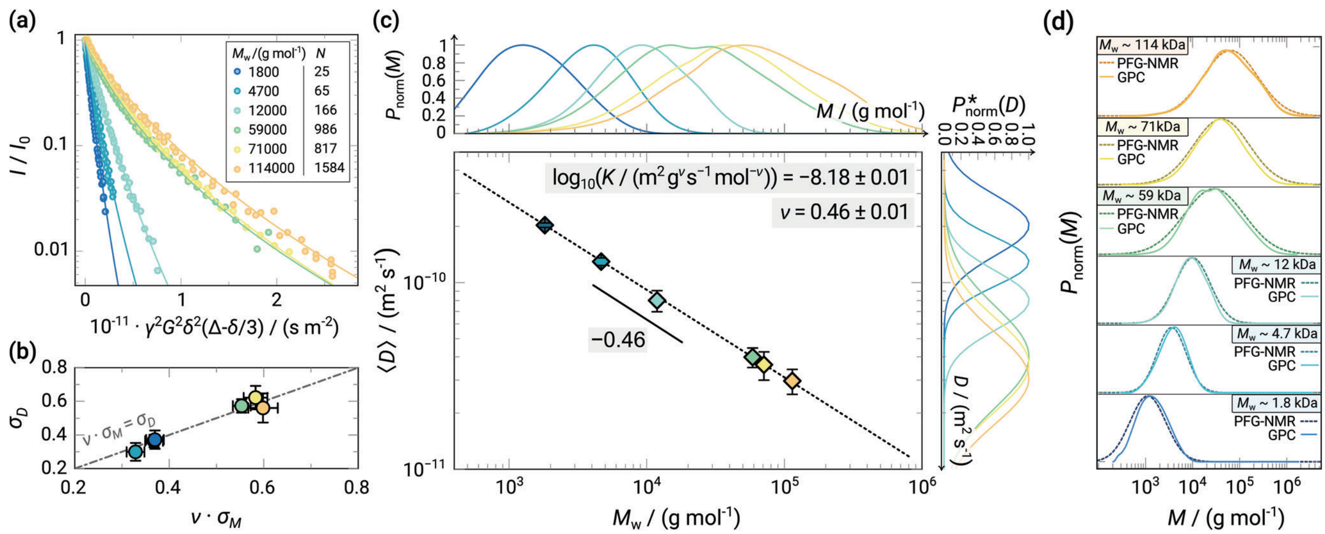

Figure 4. a) Stejskal-Tanner plot of the normalized signal decay for PAA with different M_w at pH* ≈ 1. Fitted curves are shown as solid lines. b) Standard deviations of polydispersity distributions obtained from both PFG-NMR and GPC correlated to each other. c) Scaling relation between the mean diffusion coefficient $\langle D \rangle$ from PFG-NMR and rescaled M_w from GPC for PAA. Maximum-normalized diffusion coefficient distributions $P^*_{\mathrm{norm}}(D)$ are shown on the right, GPC traces $P_{\mathrm{norm}}(M)$ on top. d) Comparison between molecular weight distributions obtained from GPC and rescaled distributions from PFG-NMR.

ing standard deviation σ_M can then be directly compared to σ_D (Figure 4b), based on the relation:

$$\sigma_D = \nu \sigma_M \tag{7}$$

The validity of this relation directly follows from the scaling properties of the log-normal distribution.[22] Finally, a polydispersity index can be determined from the diffusion coefficient distribution (see Table S2, Supporting Information) by using[12]:

$$Đ = \exp\left[\left(\sigma_D/\nu\right)^2\right] \tag{8}$$

Our analysis thus suggests that molecular weight distribution parameters of PAA, which are conventionally studied by GPC, can be reproduced by diffusion NMR to a high degree. A major advantage of the diffusional approach is that it bypasses the need for synthetic modifications of the polyelectrolyte, which is required for GPC with apolar solvents. It needs to be stressed, however, that the choice of a monomodal $P(D)$ distribution ultimately remains an assumption. While NMR-based methods are excellent to identify complex mixtures of different polymers by their chemical shift, they can hardly distinguish multiple fractions of the same polymer with similar M_w. We note that polydispersity can be well accounted for by a broad log-normal distribution, however, in case of a significant degree of asymmetry of the true $P(D)$ distribution it is not well described by the simple log-normal distribution fit model, which would lead to an artificial disagreement of the determined $\langle D \rangle$ with the true mean diffusion coefficient. While this disagreement would become more significant for higher polydispersity, it is not relevant if the $P(D)$ distribution is sufficiently symmetric with respect to the log(D) scale.

As an alternative approach to determine diffusion coefficient distributions, dynamic light scattering (DLS) can be employed. We show in the supporting information, Figure S5, Supporting Information, that DLS-derived distribution functions, polydispersities, and scaling laws are consistent with the NMR-derived results of Figure 4. However, the accuracy of DLS data is somewhat lower, in particular for low molar masses with small particle sizes.

Overall, the PFG-NMR approach presented here offers a reliable procedure to assess the molecular weight distribution for protonated PAA in a relatively short experimental time and with minimal sample preparation.

3.2.2. Diffusivity and Chain Conformation at Varying pH

So far, the diffusion at highly acidic conditions was discussed, for which polyelectrolyte effects can be safely ignored. With PFG-NMR, the diffusivity of weak polyelectrolytes can also be studied in detail when the pH value is varied. To ensure dilute conditions, weight fractions of PAA were lowered even further to 0.2 wt.%. At this concentration viscosimetric measurements of fully deprotonated PAA (114 kg mol^{-1}) at pH 9 (Figure 3) indicate a transition from semidilute polyelectrolyte behavior, for which the specific viscosity scales as $\eta_{\mathrm{sp}} \sim w_{\mathrm{PAA}}^{1/2}$, to dilute scaling behavior with $\eta_{\mathrm{sp}} \sim w_{\mathrm{PAA}}^{1}$. For further discussion, it is also necessary to convert the pH* into a degree of dissociation $\alpha = N_{\mathrm{COO^-}}/(N_{\mathrm{COO^-}} + N_{\mathrm{COOH}})$, which corresponds to the fraction of charged segments along the chain. Potentiometric titration of different PAA samples was conducted (**Figure 5**a) for the PAA samples, agreeing to previous work.[66,67] Typically, an extended Henderson-Hasselbalch equation is used to describe the dissociation curves of weak polyelectrolytes[68,69]:

$$\mathrm{p}K_\mathrm{a} = \mathrm{pH} + n \cdot \log_{10}\left(\frac{1-\alpha}{\alpha}\right) \tag{9}$$

Here, the parameter n quantifies the deviation from the pH dependence of nonpolymeric acids, in which no cooperativity

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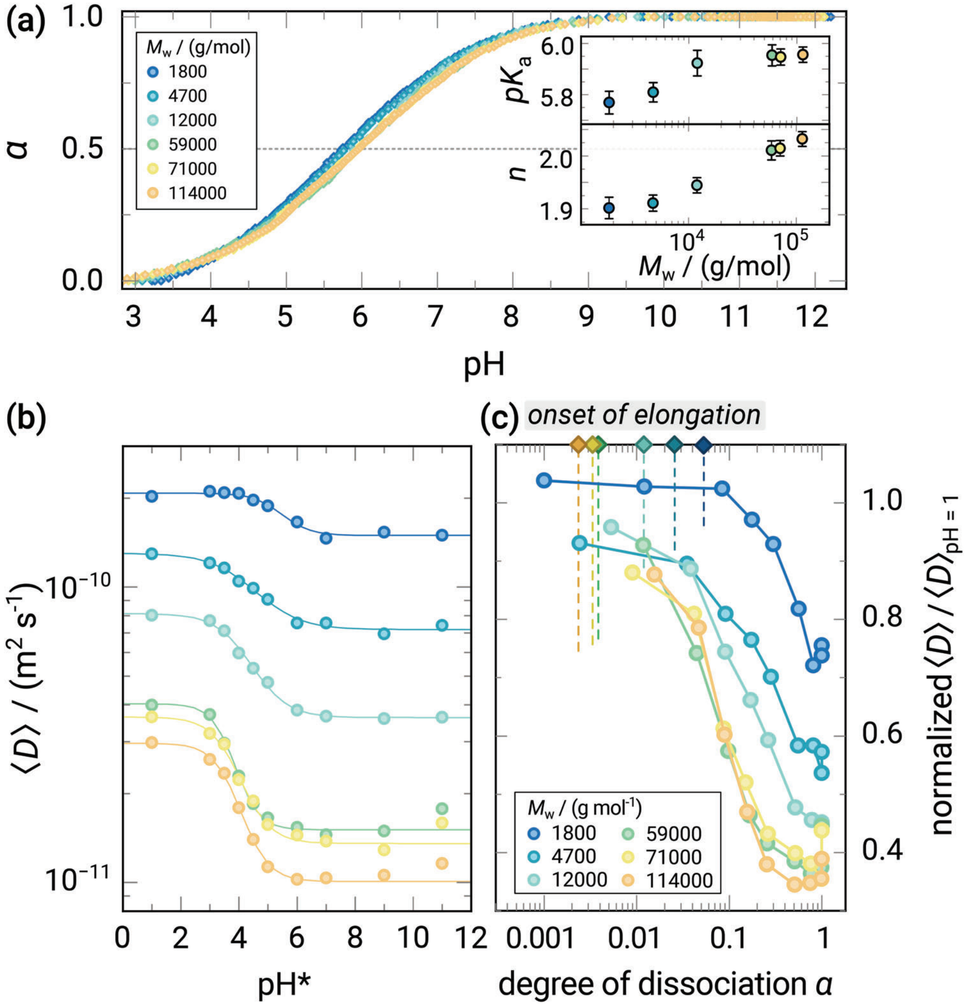

Figure 5. a) Titration curves of the investigated PAA samples in aqueous solution. The inset shows parameters obtained from fitting the extended Henderson-Hasselbalch equation (Equation (9)) to the data. b) Mean diffusion coefficients of partially neutralized PAA with different M_w for increasing pH*. (c) Mean diffusion coefficients versus degree of dissociation, normalized with respect to the value at pH* ≈ 1. The estimated onsets of coil elongation for different M_w are shown as vertical lines.

between acid sites exists.[69] In accordance with literature,[41] the M_w dependence of pK_a and n (inset of Figure 5a), indicates that cooperative effects between acid groups of PAA become stronger with increasing M_w, implying that an increasingly higher pH is required to deprotonate longer chains. At this point, a possible isotope effect on PAA dissociation in deuterated media needs to be addressed before transferring titration results to NMR data. The issue of using commercially available glass electrodes in D_2O solutions has been discussed frequently in literature: several empirical relationships pD = pH* + c of similar nature have been proposed to calculate the "true" pD (and therefore p$K_{a,D}$) value from the apparent pH* value a pH-electrode reads in D_2O solutions. Examples for values of c include 0.44,[70] 0.4,[71] or 0.41.[72] In addition, a newer study by Krężel[73] relates the pH and pH* of a compound measured in H_2O or D_2O, respectively, by an empirical formula, which we reproduced for our own instrument as pH = (0.34 ± 0.03) + (0.947 ± 0.004) · pH* (Figure S2, Supporting Information). When combining this relation with the pD/pH* conversion formula mentioned above, it becomes clear that the pH from an aqueous titration experiment is approximately the same as the pD if the experiment were to be performed in D_2O. Such a "cancel-out" approach[73] justifies using our titration data to determine the degree of dissociation α for the deuterated solutions used in NMR, though we stress that all pH* values given in this paper need to be converted to pH by the aforementioned formula if comparability with data in a protonated aqueous solvent is required.

To understand the diffusional behavior at different degrees of dissociation, diffusion coefficients for different chain lengths were measured as a function of pH* (Figure 5b). The raw data are given as Stejskal-Tanner plots of the normalized signal decay together with the respective fits in Figure S3, Supporting Information. As previously reported by Dolce as well,[36] the onset of decrease for ⟨D⟩ is size-dependent and occurs at gradually higher pH for shorter PAA chains. Consequently, less charge density along the chain is required to induce a pH-responsive change in diffusivity if the chain gets successively longer. Interestingly, this behavior contradicts Monte Carlo simulation results obtained by

Laguecir who hypothesized that longer chains require a higher persistence length and thus higher pH to expand.[41] Theoretical models of polyelectrolyte conformation however predict the opposite trend: according to the classical Flory-like approach for example,[74,75] both entropic chain elasticity and electrostatic repulsion contribute to the free energy of a spherical polyelectrolyte chain with uniform charge distribution[76]:

$$F_{\mathrm{Flory}}(R) = k_{\mathrm{B}} T \left[\frac{R^2}{Nb^2} + \frac{l_{\mathrm{B}} (N\alpha)^2}{R} \right] \tag{10}$$

Here, l_{B} represents the Bjerrum length, and b is the length of a chain segment. The latter electrostatic interaction term in Equation (10) makes the polymer adopt an elongated rodlike shape, hence increasing the effective hydrodynamic radius. Taking the swelling into account, the chain size along the elongated axis then scales with chain size and degree of charge in a nonlinear manner by $R_{\mathrm{e}} \sim N\alpha^{2/3}[l_{\mathrm{B}} b^2 \ln(N)]^{1/3} R_{\mathrm{e}} \sim N\alpha^{2/3}[l_{\mathrm{B}} b^2$.[75,76]

A theoretical prediction for the onset of chain elongation is given by a critical degree of dissociation α_{crit} at which the electrostatic energy of the charged chain corresponds to the thermal energy.[75] Above this critical value, electrostatic repulsion starts to affect the shape of the polymer. For ideal chain statistics, Flory theory predicts[76]:

$$\alpha_{\mathrm{crit}} \approx \sqrt{\frac{b}{l_{\mathrm{B}}}} N^{-\frac{3}{4}} \tag{11}$$

where the Bjerrum length at 25 °C was calculated as $l_{\mathrm{B}} = e^2/(4\pi\epsilon\epsilon_0 k_{\mathrm{B}} T) = 0.715$ nm and the monomer length $b \approx 2 l_{\mathrm{C-C}} \sin(\varphi_{\mathrm{C-C}}/2) = 2 \cdot 0.152\ \mathrm{nm} \cdot \sin(109.47°/2) = 0.248$ nm was geometrically estimated.[77] When comparing this theoretical critical degree of charge with the α-dependent, normalized diffusion coefficients (Figure 5c) for short chains, it becomes apparent that large changes in D only occur beyond the critical value α_{crit}. Thus, for short chains, the Flory theory criterion is suitable to estimate the onset of chain elongation. For longer chains, however, α_{crit} becomes too small to be even comparable to experimental values determined by titration, which become unreliable for $\alpha < 0.01$. Elongational changes of the conformation are hence already expected for extraordinarily small degrees of dissociation. In the range of α which is accessible for titrations, the experimental data are, however, in agreement with chain elongation of long chains starting already at very low dissociation, see Figure 5c.

In agreement with measurements by Dolce and Mériguet,[36] we observe that *no* threshold chain length $M_{\mathrm{n}} < 16.5$ kDa, as postulated by Swift et al.,[35] exists for the pH dependence of PAA diffusion. All molecular weights in our work (>1.8 kDa) show a pronounced decrease of $\langle D \rangle$ with increasing dissociation. Poor comparability between the different studies might however result from different concentration regimes being investigated: the PAA weight fractions used for DOSY measurements by Swift et al. seem to lie in the semidilute range, which, as the authors remark, could lead to significant chain overlap. In Dolce's investigation,[36] chain overlap was instead ruled out by concentration-dependent diffusion measurements. In order to explain why Swift et al. observed a discontinuity in their pH-dependent diffusion data for low chain lengths, they hypothesized that for very short chains, it becomes increasingly unlikely for a monomer to form hydrogen bonds to a side group far away along the chain. In that case, the conformation of chains with higher M_{w} is more sensitively affected by the disruption of hydrogen bonding which results from an increased swelling behavior in pH.[35] However, this proposed effect is not evident in our data, which suggests that even for short-chain PAA, diffusivity is strongly dependent on pH.

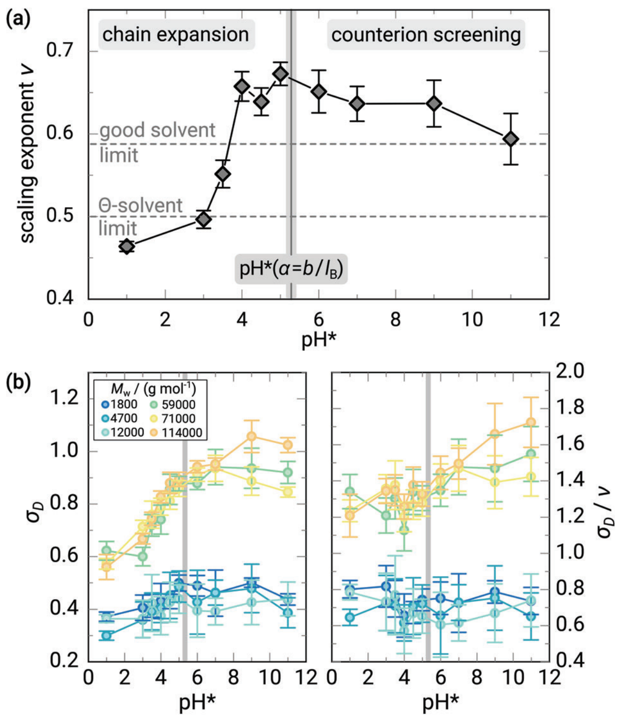

Figure 6. a) Scaling exponents in dilute PAA solutions at various apparent pH*. b) Standard deviations of the log-normal diffusion coefficient distribution for PAA with varying M_{w} versus pH* (left) and the same data rescaled by the scaling exponent ν (right).

Just like previously outlined for acidic conditions, the scaling exponent in Equation (6) can now be tracked as a function of the pH in order to study how chain conformation evolves (**Figure 6**a, see also Figure S4 and Table S3, Supporting Information, for fits to the raw data and parameters).

After exceeding the good-solvent limit at pH* ≈ 3.7, the observed scaling exponent ν reaches a maximum at pH* = 5, before it levels off and even slightly decreases towards higher pH* values. The maximum point coincides well with the pH* at which the degree of dissociation is equivalent to $\alpha = b/l_{\mathrm{B}} \approx 0.347$, as predicted by the Oosawa–Manning model.[78,79] Here, the counterion condensation criterion for the Manning parameter $\xi = l_{\mathrm{B}}/b_{\mathrm{el}} \geq 1$ is fulfilled, where $b_{\mathrm{el}} = b/\alpha$ is the distance between two charged groups along the chain.

Such a shift in scaling behavior therefore indicates two opposing effects: the electrostatic repulsion initially induces chain expansion with increasing pH, causing ν to rise. As the counterion screening becomes increasingly significant with further addition of base, the persistence length decreases and the chain readopts a less expanded chain dimension. In fact, experimental evidence for PAA solutions undergoing a transition of conformational behavior around this point is numerous, including changes in the water structure around PAA shown by Raman spectroscopy,[80] a maximum in the reduced viscosity,[81] a change in slope for the intrinsic viscosity[82] or the refractive index[83] or a maximum of electrical birefringence.[84,85] The apparent loss in solvent quality in this regime was equivalently demonstrated by Schweins et al. in light scattering experiments of sodium polyacrylate solutions at pH = 9 upon addition of NaCl.[37] In our present work, we can clearly attribute these more or less empirical results to the molecular structure, namely the chain conformation. Notably, the rigid rod limit $\nu = 1$, as it can be experimentally observed for the strong polyelectrolyte NaPSS,[43,45,47] is not observed for PAA, which retains some degree of flexibility in its highly charged state.

We note a further subtlety in the data, namely that at high pH the diffusion coefficients of long chains start to increase again with pH, see Figure 5b, while the scaling exponent decreases (Figure 6a), which can be explained by a different screening behavior. Figure S6a,b, Supporting Information, shows the ionic strengths and a comparison of the hydrodynamic radius to the Debye length, indicating that long chains are fully screened, while short chains are not. The influence on the scaling laws, see Figure S4, Supporting Information, is however minimal and two different scaling regimes can hardly be identified.

Changes in behavior further become apparent when comparing the width of the diffusion coefficient distribution σ_D at different pH* (Figure 6b), which reflects the variation of the chain ensemble diffusivity. A collection of chains will in principle display the same molecular weight distribution $P(M)$ regardless of pH. As predicted by Equation (7) however, an increase in the scaling exponent ν should be accompanied by an inevitable broadening of the observed $P(D)$. Accordingly, the standard deviation σ_D obtained from PFG-NMR indeed displays a steady increase with pH*, in particular for longer chains. The rescaled quantity σ_D/ν should remain constant on the other hand, as the underlying $P(M)$ distribution stays unchanged. Despite a large error, this is approximately the case for up to pH* = 5. Beyond this point, however, σ_D/ν continues to increase for the high M_w samples, which cannot be explained by Equation (7) alone. The occurrence of counterion screening, induced by titration with the base, thus seems to impart an additional degree of conformational broadening on the chain ensemble. By means of Monte Carlo simulations, Laguecir et al. had argued with counteracting effects of an increase of ionic strength at constant pH: On the one hand, ionic strength would lead to an increase in the degree of dissociation, inducing an increase in the radius of gyration.[41] On the other hand, salt screening facilitates dissociation of the acid groups, the reduced Debye length counteracts electrostatic repulsion along the chain and reduces the radius of gyration.[41] We can argue here, that the additional degrees of freedom induced by the counterion distribution might allow the system to explore a larger variety of possible chain conformations, which acts as an additional mechanism of broadening for the distribution of the radius of gyration, and thus the distribution of diffusion coefficients.

4. Conclusion

Self-diffusion coefficients of PAA at various pH* have been measured and related to molecular weights by scaling laws using pulsed field gradient NMR. The scaling exponent $\nu = 0.47$ indicates a slightly collapsed conformation at pH* ≈ 1. By fitting a log-normal distribution-based model to signal decay data, diffusion coefficient distributions were extracted. These were rescaled to obtain molecular weight distributions, showing excellent agreement with GPC results. Although the presented method is limited to resolve monomodal molecular weight distributions, it was demonstrated that PFG-NMR in combination with the presented fitting procedure proves to be a powerful tool to estimate molecular weights and polydispersites of PAA.

When systematically varying the pH*, an increasing degree of dissociation induces a major drop in self-diffusivity of PAA, which occurs at lower pH* for longer chains. Our analysis confirms that this behavior remains universally valid down to a chain length $N = 25$, refuting the notion of a critical lower molar mass for the pH-responsiveness of PAA chain conformations. The scaling exponent indicates a maximum chain expansion at pH* = 5 ($\nu = 0.67$), before counterion screening makes the charged chain readopt a less expanded conformation at even higher pH*. In the basic regime, where counterion screening becomes relevant, the diffusion coefficient distribution appears broader than expected, which we explain by a broader variety of accessible conformations, implied by the larger degrees of freedom with respect to the spatial distribution of counterions.

5. Experimental Section

Materials: Poly(acrylic acid) (PAA) was commercially purchased from different vendors ($M_w \sim$ 2 000, 100 000, 250 000, all in g mol^{-1}: Sigma Aldrich; 50 000 g mol^{-1}: PolyScience Inc.). Polymers which were purchased as sodium polyacrylate salts ($M_w \approx$ 5 100, 15 000, all in g mol^{-1}: Sigma Aldrich) were first fully protonated by dialysis (Nalo Cellulose, Carl Roth, Karlsruhe, pore size 25 Å to 30 Å) in dilute HCl for 3 days and subsequently purified by dialysis in water for further 5 days with frequent exchange of the dialysis medium. The solid PAA was then obtained by freeze-drying. All aqueous solutions were prepared with water purified by a Milli-Q system (resistivity 18.2 MΩ cm^{-1}). Deuterated substances (all from Sigma Aldrich) included D_2O (99.9 atom% D), NaOD (39.7 wt.% in D_2O, 99.7 atom% D), DCl (35 wt.% in D_2O, 99.9 atom% D). The latter two were diluted with D_2O to obtain appropriately concentrated stock solutions.

Modification of PAA to Poly(benzyl acrylate): Dry poly(acrylic acid) was dissolved in dimethyl sulfoxide (DMSO), which was dried over a molecular sieve (4 Å) beforehand, in order to obtain solutions with ≈ 10 wt.% polymer. In a glass vial, 1,1,3,3-tetramethylguanidine (2 equivalents, Sigma Aldrich, 99 %) was added dropwise to the solution, after which gradual neutralization of the PAA led to a sharp rise of the viscosity. After stirring for 5 min the mixture appeared homogeneous. Then, benzyl bromide (1.5 equivalents, Sigma Aldrich, 98%) was slowly added dropwise while cooling the reaction mixture in an ice bath just above the melting point of DMSO. The reaction mixture was purged with Ar and left stirring for 24 h at room temperature in closed vials. Afterward, the benzylated polymer was precipitated by slowly pipetting the solution into a large excess of ice-cold dilute acetic acid and subsequently filtered to obtain a white solid. The poly(benzyl acrylate) (PBzA) was further purified by dialysis (Nalo Cellulose, pore size 25 Å to 30 Å) in frequently exchanged water at room temperature for 5 days and subsequent drying in vacuum (40 mPa) at 50 °C.

Potentiometric Titration and pH Adjustment: The degree of charge of PAA as a function of pH was determined by potentiometric titration with an inoLab pH Level 1 pH meter (WTW, Xylem Analytics, Weilheim) equipped with a SenTix Mic electrode (WTW, Xylem Analytics, Weilheim). PAA was dissolved in water to obtain a 0.07 M polymer solution (referring to the monomer concentration) and titrated against 0.1 M NaOH while recording the pH. A gas flow of Ar was directed above the stirred titrand solution to prevent the dissolution of CO_2. The degree of dissociation was calculated from the titrated volume V_{NaOH} as $\alpha = V_{NaOH}$ / V_{eq} under the assumption that all chains are protonated before titration. Here, the equivalence point volume V_{eq} was determined from the maximum of the numerical derivative dpH / dV_{NaOH}. For pH-dependent diffusion measurements, the pH* value (apparent pH value measured for D_2O solutions) was adjusted by the addition of a 0.5 M NaOD solution to a 0.2 wt.% PAA solution in D_2O. 300 µL aliquots for NMR-measurements were taken at specific pH* values.

Gel Permeation Chromatography: All measurements were performed on a 1260 Infinity instrument (Polymer Standard Service, Mainz) equipped with a refractive index detector. Dimethylacetamid with 50 mM LiBr was used as eluent for the three GRAM columns (polyester particles) at 70 °C and a flow rate of 1 mL min^{-1}. The samples were prepared by dissolving 2 mg poly(benzyl acrylate) in the eluent and passing it through a PTFE syringe filter (0.2 µm). The system was calibrated using poly(styrene) and poly(methyl methacrylate) standards (Polymer Standard Service, Mainz) and using the supplied software WinGPC UniChrom. Based on reported Hildebrandt solubility parameters[86,87] for PBzA ($\delta = 19.5$ MPa$^{1/2}$), the PMMA standard ($\delta = 18.6$ MPa$^{1/2}$) was chosen over the PS standard ($\delta = 17.6$ MPa$^{1/2}$) to determine M_w values.

Nuclear Magnetic Resonance Spectroscopy: A 400 MHz NMR spectrometer (AVANCE III HD 400, Bruker BioSpin, Ettlingen, Germany), equipped with a gradient probe head (DiffBBO, Bruker BioSpin) was used for NMR measurements at 25 °C. Signal attenuation of the polymer peak was observed using a stimulated echo pulse sequence (Figure 1b), where the gradient strength G was varied from 0.2 T m^{-1} up to a maximum value ranging from 3 T m^{-1} to 6 T m^{-1}, according to the sample. The gradient pulse duration and diffusion time were set to $\delta = 1$ ms and $\Delta = 100$ ms, respectively. The echo time between first and second radiofrequency pulse was $\tau_{echo} = 2.11$ ms. Raw data were processed in Topspin 3.6.3 (Bruker BioSpin) and fits were conducted using MATLAB R2015b (MathWorks, Natick). At pH* $\approx$ 1, measurements were reproduced for at least 3 independently prepared solutions and for different maximum values of the gradient strength. The decay data from each measurement were combined before fitting. For evaluation of the decay of the backbone protons peaks, the chemical shift region from 2.8 ppm to 1.4 ppm was integrated. A signal decay of at least one order of magnitude was required for reliable fitting, which was performed using the optimization function lsqnonlin from MATLAB to conduct a least square fit to a two-parameter model according to Equation (4).

Viscometry: Viscosities at 25 °C were determined by rotational rheometry, using an MCR 101 rheometer (Anton Paar, Ostfildern) equipped with a CP50-2 cone plate configuration (plate diameter 50 mm, gap size 213 µm, cone angle 2°) and a Peltier element for temperature control. The shear rate $\dot{\gamma}$ was varied from 10 s^{-1} to 600 s^{-1} and in case of shear thinning the zero shear viscosity was determined from the viscosity plateau at low $\dot{\gamma}$.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

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[1] D. I. Bower, *An Introduction to Polymer Physics*, 1 ed., Cambridge University Press, Cambridge **2002**.
[2] D. Berek, *J. Sep. Sci.* **2010**, *33*, 315.
[3] J. M. G. Cowie, V. Arrighi, *Polymers: Chemistry and Physics of Modern Materials*, 3rd ed., CRC Press, Boca Raton, Fla **2007**.
[4] I. Lacík, S. Beuermann, M. Buback, *Macromolecules* **2001**, *34*, 6224.
[5] R. J. Minari, G. Caceres, P. Mandelli, M. M. Yossen, M. Gonzalez-Sierra, J. R. Vega, L. M. Gugliotta, *Macromol. React. Eng.* **2011**, *5*, 223.
[6] J. Loiseau, N. Doërr, J. M. Suau, J. B. Egraz, M. F. Llauro, C. Ladavière, J. Claverie, *Macromolecules* **2003**, *36*, 3066.
[7] F.-D. Kuchta, A. M. Van Herk, A. L. German, *Macromolecules* **2000**, *33*, 3641.
[8] L. Couvreur, C. Lefay, J. Belleney, B. Charleux, O. Guerret, S. Magnet, *Macromolecules* **2003**, *36*, 8260.
[9] I. Chaduc, A. Crepet, O. Boyron, B. Charleux, F. D'agosto, M. Lansalot, *Macromolecules* **2013**, *46*, 6013.
[10] I. Lacík, M. Stach, P. Kasák, V. Semak, L. Uhelská, A. Chovancová, G. Reinhold, P. Kilz, G. Delaittre, B. Charleux, I. Chaduc, F. D'agosto, M. Lansalot, M. Gaborieau, P. Castignolles, R. G. Gilbert, Z. Szablan, C. Barner-Kowollik, P. Hesse, M. Buback, *Macromol. Chem. Phys.* **2015**, *216*, 23.
[11] Q. Li, Y. Bao, H. Wang, F. Du, Q. Li, B. Jin, R. Bai, *Polym. Chem.* **2013**, *4*, 2891.
[12] M. Röding, D. Bernin, J. Jonasson, A. Särkkä, D. Topgaard, M. Rudemo, M. Nydén, *J. Magn. Res.* **2012**, *222*, 105.
[13] P. Stilbs, *Diffusion and Electrophoretic NMR*, De Gruyter, Berlin, Boston **2019**.
[14] X. Guo, E. Laryea, M. Wilhelm, B. Luy, H. Nirschl, G. Guthausen, *Macromol. Chem. Phys.* **2017**, *218*, 1600440.
[15] M. Foroozandeh, L. Castañar, L. G. Martins, D. Sinnaeve, G. D. Poggetto, C. F. Tormena, R. W. Adams, G. A. Morris, M. Nilsson, *Angew. Chem.* **2016**, *128*, 15808.
[16] G. Peat, P. J. Boaler, C. L. Dickson, G. C. Lloyd-Jones, D. Uhrín, *Nat. Commun.* **2023**, *14*, 4410.
[17] W. Li, H. Chung, C. Daeffler, J. A. Johnson, R. H. Grubbs, *Macromolecules* **2012**, *45*, 9595.
[18] W. S. Price, *NMR Studies of Translational Motion*, Cambridge University Press, Cambridge **2010**.
[19] Y. Cohen, L. Avram, L. Frish, *Angew. Chem., Int. Ed.* **2005**, *44*, 520.
[20] G. Fleischer, D. Geschke, J. Kärger, W. Heink, *J. Magn. Reson.* **1985**, *65*, 429.
[21] H. Walderhaug, O. Söderman, D. Topgaard, *Prog. Nucl. Magn. Reson. Spectrosc.* **2010**, *56*, 406.
[22] M. Röding, S. J. Bradley, N. H. Williamson, M. R. Dewi, T. Nann, M. Nydén, *PLoS One* **2016**, *11*, e0155718.
[23] P. T. Callaghan, D. N. Pinder, *Macromolecules* **1981**, *14*, 1334.
[24] A. Chen, D. Wu, C. S. Johnson, Jr., *J. Am. Chem. Soc.* **1995**, *117*, 7965.
[25] B. Håkansson, M. Nydén, O. Söderman, *Colloid Polym. Sci.* **2000**, *278*, 399.
[26] J.-G. Rosenboom, J. De Roo, G. Storti, M. Morbidelli, *Macromol. Chem. Phys.* **2017**, *218*, 1600436.
[27] N. H. Williamson, M. Röding, S. J. Miklavcic, M. Nydén, *J. Colloid Interface Sci.* **2017**, *493*, 393.
[28] X. Gong, E. W. Hansen, Q. Chen, *Macromol. Chem. Phys.* **2011**, *212*, 1007.

[29] P. T. Callaghan, D. N. Pinder, *Macromolecules* **1985**, *18*, 373.
[30] E. Ruzicka, P. Pellechia, B. C. Benicewicz, *Anal. Chem.* **2023**, *95*, 7849.
[31] N. E. Kuz'mina, S. V. Moiseev, V. I. Krylov, V. A. Yashkir, V. A. Merkulov, *J. Anal. Chem.* **2014**, *69*, 953.
[32] N. E. Kuz'mina, S. V. Moiseev, V. I. Krylov, V. A. Yashkir, V. A. Merkulov, *J. Anal. Chem.* **2015**, *70*, 843.
[33] P. T. Callaghan, J. Lelievre, *Biopolymers* **1985**, *24*, 441.
[34] M. Nydén, O. Söderman, *Macromolecules* **1998**, *31*, 4990.
[35] T. Swift, L. Swanson, M. Geoghegan, S. Rimmer, *Soft Matter* **2016**, *12*, 2542.
[36] C. Dolce, G. Mériguet, *Colloid Polym. Sci.* **2017**, *295*, 279.
[37] R. Schweins, J. Hollmann, K. Huber, *Polymer* **2003**, *44*, 7131.
[38] D. F. Anghel, V. Alderson, F. M. Winnik, M. Mizusaki, Y. Morishima, *Polymer* **1998**, *39*, 3035.
[39] D. G. Mintis, V. G. Mavrantzas, *J. Phys. Chem. B* **2019**, *123*, 4204.
[40] A. K. Gupta, U. Natarajan, *Mol. Simul.* **2019**, *45*, 876.
[41] A. Laguecir, S. Ulrich, J. Labille, N. Fatin-Rouge, S. Stoll, J. Buffle, *Eur. Polym. J.* **2006**, *42*, 1135.
[42] J. Landsgesell, L. Nová, O. Rud, F. Uhlík, D. Sean, P. Hebbeker, C. Holm, P. Kosovan, *Soft Matter* **2019**, *15*, 1155.
[43] M. G. Oostwal, M. H. Blees, J. De Bleijser, J. C. Leyte, *Macromolecules* **1993**, *26*, 7300.
[44] M. Sedlák, E. J. Amis, *J. Chem. Phys.* **1992**, *96*, 817.
[45] C. G. Lopez, J. Linders, C. Mayer, W. Richtering, *Macromolecules* **2021**, *54*, 8088.
[46] R. De, D. Ray, B. Das, *RSC Adv.* **2015**, *5*, 54890.
[47] S. K. Filippov, T. A. p Seery, J. KriZ, M. Hruby, P. Cernoch, O. SedlaCek, P. Kadlec, J. Pánek, P. Stepánek, *Polym. Int.* **2013**, *62*, 1271.
[48] M. Sedlák, *Colloid Polym. Sci.* **2017**, *295*, 1281.
[49] A. A. Istratov, O. F. Vyvenko, *Rev. Sci. Instrum.* **1999**, *70*, 1233.
[50] A. Scotti, W. Liu, J. S. Hyatt, E. S. Herman, H. S. Choi, J. W. Kim, L. A. Lyon, U. Gasser, A. Fernandez-Nieves, *J. Chem. Phys.* **2015**, *142*, 234905.
[51] M. Baumgaertel, H. H. Winter, *Rheol. Acta* **1989**, *28*, 511.
[52] J. Honerkamp, J. Weese, *Macromolecules* **1989**, *22*, 4372.
[53] K. P. Whittall, A. L. MacKay, *J. Magn. Res.* **1989**, *84*, 134.
[54] K. P. Whittall, M. J. Bronskill, R. Henkelman, *J. Magn. Reson.* **1991**, *95*, 221.
[55] M. Röding, N. H. Williamson, M. Nydén, *J. Magn. Res.* **2015**, *261*, 6.
[56] N. H. Williamson, M. Nydén, M. Röding, *J. Magn. Reson.* **2016**, *267*, 54.
[57] S. W. Provencher, *Comput. Phys. Commun.* **1982**, *27*, 213.
[58] I. J. Day, *J. Magn. Reson.* **2011**, *211*, 178.
[59] K. F. Morris, C. S. Johnson, *J. Am. Chem. Soc.* **1992**, *114*, 3139.
[60] D. Bernin, D. Topgaard, *Curr. Opin. Colloid Interface Sci.* **2013**, *18*, 166.
[61] R. H. Colby, *Rheol. Acta* **2010**, *49*, 425.
[62] W.-M. Kulicke, C. Clasen, *Viscosimetry of Polymers and Polyelectrolytes*, Springer, Berlin, Heidelberg **2004**.
[63] R. P. Choudhury, P. Galvosas, M. Schönhoff, *J Phys Chem B* **2008**, *112*, 13245.
[64] B. Li, L. Xu, Q. Wu, T. Chen, P. Sun, Q. Jin, D. Ding, X. Wang, G. Xue, A.-C. Shi, *Macromolecules* **2007**, *40*, 5776.
[65] J. Dong, Y. Ozaki, K. Nakashima, *Macromolecules* **1997**, *30*, 1111.
[66] M. Tekaat, D. Bütergerds, M. Schönhoff, A. Fery, C. Cramer, *Phys. Chem. Chem. Phys.* **2015**, *17*, 22552.
[67] D. Bütergerds, C. Kateloe, C. Cramer, M. Schönhoff, *J. Polym. Sci., Part B: Polym. Phys.* **2017**, *55*, 425.
[68] A. Katchalsky, P. Spitnik, *J. Polym. Sci.* **1947**, *2*, 432.
[69] A. I. Petrov, A. A. Antipov, G. B. Sukhorukov, *Macromolecules* **2003**, *36*, 10079.
[70] K. Mikkelsen, S. O. Nielsen, *J. Phys. Chem.* **1960**, *64*, 632.
[71] P. K. Glasoe, F. A. Long, *J. Phys. Chem.* **1960**, *64*, 188.
[72] A. K. Covington, M. Paabo, R. A. Robinson, R. G. Bates, *Anal. Chem.* **1968**, *40*, 700.
[73] A. Krężel, W. Bal, *J. Inorg. Biochem.* **2004**, *98*, 161.
[74] W. Kuhn, O. Künzle, A. Katchalsky, *Helv. Chim. Acta* **1948**, *31*, 1994.
[75] A. Dobrynin, M. Rubinstein, *Prog. Colloid Polym. Sci.* **2005**, *30*, 1049.
[76] Barrat, J.-L.; Joanny, J.-F., *Advances in Chemical Physics: Theory of Polyelectrolyte Solutions* (Eds: I. Prigogine, S. A. Rice), John Wiley & Sons, Inc, New York **1996**.
[77] W. M. Haynes, D. R. Lide, T. J. Bruno, *CRC Handbook of Chemistry and Physics*, 97th ed., CRC Press, Boca Raton, Fla **2016**.
[78] F. Oosawa, *Polyelectrolytes*, Marcel Dekker, Inc., New York **1971**.
[79] G. S. Manning, *J. Chem. Phys.* **1969**, *51*, 924.
[80] N. Tsukida, H. Muranaka, M. Ide, Y. Maeda, H. Kitano, *J. Phys. Chem. B* **1997**, *101*, 6676.
[81] R. Sakamoto, K. Imahori, *Nippon Kagaku Zasshi* **1962**, *83*, 389.
[82] G. Staikos, G. Bokias, *Polym. Int.* **1993**, *31*, 385.
[83] A. Ikegami, *J. Polym. Sci., Part A: Gen. Pap.* **1964**, *2*, 907.
[84] H. Hoffmann, K. Kamburova, H. Maeda, T. Radeva, *Colloids Surf., A* **2010**, *354*, 61.
[85] H. Hoffmann, M. Liria Turco Liveri, F. Paolo Cavasino, *J. Chem. Soc. Faraday Trans.* **1997**, *93*, 3161.
[86] H. Ahmad, M. Yaseen, *Polym. Eng. Sci.* **1979**, *19*, 858.
[87] C. M. Hansen, *Hansen solubility parameters: A user's handbook*, 2nd ed., CRC, Boca Raton, Florida **2007**.